

SPANGDAHLE AB, Germany

WMO: 106070

Lat: 49.973N

Lon: 6.693E

Elev: 1197

StdP: 14.07

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	20.9	10.5	9.8	21.5	15.5	12.6	23.1	27.0	44.5	24.5	44.3	7.6	70	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	85.6	66.8	81.5	65.1	78.3	63.9	68.7	80.5	67.1	78.0	65.3	74.9	8.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	96.0	71.6	63.0	90.1	69.9	61.9	86.5	68.9	33.6	80.2	32.2	77.4	30.8	75.1	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.9	16.8	DB	11.9	91.7	5.2	3.3	8.2	94.1	5.2	96.1	2.3	97.9	-1.5	100.3	(4)
(5)			WB	10.6	72.2	5.1	2.3	7.0	73.9	4.0	75.2	1.1	76.5	-2.5	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	34.3	36.1	41.9	48.8	55.7	61.2	64.7	64.0	57.2	49.6	41.1	35.7	(6)	
	DBStd	12.63	7.23	6.98	6.11	7.22	6.63	6.19	6.25	5.63	5.39	6.39	6.73	7.06	(7)	
	HDD50	2072	486	388	258	107	21	1	0	0	5	87	275	444	(8)	
	HDD65	5919	951	808	716	485	296	147	84	87	240	479	718	908	(9)	
	CDD50	1808	0	0	7	73	198	337	457	434	222	73	7	0	(10)	
	CDD65	182	0	0	0	1	8	34	76	56	7	0	0	0	(11)	
	CDH74	1791	0	0	0	13	87	338	724	547	80	2	0	0	(12)	
	CDH80	488	0	0	0	0	12	86	225	153	12	0	0	0	(13)	
	WSAvg	7.8	9.1	8.7	8.9	7.7	7.5	7.2	7.1	6.5	6.8	7.4	7.8	8.6	(14)	
	PrecAvg	30.2	2.8	2.2	2.0	1.9	2.6	2.8	2.8	2.6	2.4	2.6	2.6	3.1	(15)	
(16) Precipitation	PrecMax	39.7	7.8	5.9	5.6	4.8	5.0	6.7	7.6	5.7	5.4	7.2	5.2	8.5	(16)	
	PrecMin	22.2	0.3	0.2	0.3	0.0	0.5	0.8	0.9	0.4	0.7	0.9	0.2	0.4	(17)	
	PrecStd	4.3	1.5	1.3	1.3	1.0	1.2	1.5	1.4	1.3	1.4	1.3	1.2	1.7	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	55.0	66.0	75.7	81.0	87.7	91.4	90.1	82.0	71.7	59.1	52.1	(19)	
		MCWB	49.4	46.2	52.0	57.4	64.4	68.1	68.2	68.5	64.3	60.0	55.3	50.4	(20)	
	2%	DB	49.8	50.1	60.5	71.3	75.8	82.2	86.0	84.2	75.5	65.9	55.2	50.0	(21)	
		MCWB	47.9	45.3	49.4	55.6	61.2	65.9	66.9	66.2	62.8	58.8	52.3	48.3	(22)	
	5%	DB	46.7	47.9	55.5	66.3	71.9	77.4	81.9	80.2	71.4	61.2	52.8	47.8	(23)	
		MCWB	45.4	45.0	47.0	53.8	59.0	63.7	65.3	64.6	60.8	56.5	50.5	46.6	(24)	
	10%	DB	44.4	45.5	51.7	62.1	68.1	73.6	77.5	76.0	67.1	58.8	50.3	45.0	(25)	
		MCWB	43.2	43.7	46.1	51.7	57.7	62.1	63.6	63.3	59.0	55.3	48.4	44.1	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	49.3	53.2	60.0	67.1	71.1	70.9	70.3	67.1	62.0	56.1	51.3	(27)	
		MCDB	51.6	51.5	61.8	70.8	76.7	81.4	85.2	82.6	77.4	67.6	57.5	51.6	(28)	
	2%	WB	48.2	46.9	50.4	57.2	63.8	67.8	68.6	68.1	64.1	59.7	53.1	48.5	(29)	
		MCDB	49.2	48.6	57.0	68.0	71.9	77.9	81.6	80.2	73.0	64.5	55.0	49.5	(30)	
	5%	WB	45.5	45.3	48.9	54.8	61.1	65.5	66.9	66.4	62.0	57.6	51.0	46.5	(31)	
		MCDB	46.6	47.0	53.9	64.1	68.8	75.2	78.7	76.1	69.3	61.1	52.9	47.4	(32)	
	10%	WB	42.8	43.5	46.9	52.7	58.7	63.3	65.0	64.6	60.0	55.6	48.9	44.3	(33)	
		MCDB	43.7	45.2	50.9	60.6	66.1	71.8	75.0	73.4	65.8	58.4	50.5	45.2	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	7.9	10.1	14.0	17.2	17.4	17.5	18.2	18.3	15.9	12.9	8.5	7.6	(35)	
		MCDBR	9.9	14.0	22.0	24.1	23.6	24.2	25.2	25.8	22.4	17.2	10.8	9.3	(36)	
		MCWBR	8.8	10.4	12.8	12.1	11.6	11.6	10.1	11.1	11.5	11.2	8.2	8.5	(37)	
	5% WB	MCDBR	9.3	11.9	17.6	21.2	20.6	21.3	22.1	21.6	19.5	15.5	9.9	8.9	(38)	
		MCWBR	8.7	9.6	11.4	11.4	11.1	11.5	9.9	10.8	10.9	10.7	8.4	8.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.330	0.361	0.392	0.400	0.403	0.401	0.397	0.368	0.352	0.325	0.302	(40)	
	taud		2.433	2.382	2.315	2.264	2.271	2.285	2.283	2.323	2.381	2.424	2.434	2.423	(41)	
	Ebn at Noon		236	256	266	270	272	272	270	266	261	245	224	218	(42)	
	Edh at Noon		20	26	33	39	41	40	40	36	31	25	20	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		249	476	846	1334	1572	1744	1653	1417	1059	612	282	193	(44)	
	RadStd		38	90	129	164	157	159	180	144	129	59	34	29	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page